Gradient Descent

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Gradient descent

Definition 0.1 (Gradient descent)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ have first partial derivatives, let $\underset{\sim}{x}^{(0)}$ be a starting point, and let $s^{(0)}, s^{(1)}, \ldots$ be positive **step lengths**. **Gradient descent** [1] computes, for $t = 0, 1, 2, \ldots$,

$$\underset{\sim}{x}^{(t+1)} \stackrel{\text{def}}{=} \underset{\sim}{x}^{(t)} - s^{(t)} \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^{(t)}).$$

Example 0.1 (Fixed steps on a parabola)**.** For $f(x) = x^2$, $f'(x) = 2x$, so gradient descent with fixed step length s is

$$\begin{aligned} x^{(t+1)} &= x^{(t)} - 2s\, x^{(t)} \\ &= (1 - 2s)\, x^{(t)}, \end{aligned}$$

and $x^{(t)} = (1 - 2s)^t x^{(0)}$.

- With $s = 0.25$, $x^{(t)} = 0.5^t x^{(0)}$, which tends to the minimizer 0.
- With $s = 0.5$, the first step lands on 0, since $1 - 2s = 0$.
- With $s = 1.5$, $x^{(t)} = (-2)^t x^{(0)}$, which moves away from 0 for any $x^{(0)} \neq 0$: too long a step can make gradient descent fail.

Remark (Gradient descent as Newton's method with a simpler Hessian)

Remark 0.1 (Gradient descent as Newton's method with a simpler Hessian). Gradient descent (Definition 0.1) is Newton's method (Definition 5 in Newton's Method) with the Hessian replaced by $\frac{1}{s^{(t)}} \mathbf{I}_p$, so it needs no second derivatives.

Definition 0.2 (Gradient ascent)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ have first partial derivatives, let $\underset{\sim}{x}^{(0)}$ be a starting point, and let $s^{(0)}, s^{(1)}, \ldots$ be positive step lengths. **Gradient ascent** [1] repeats, for $t = 0, 1, 2, \ldots$, the update

$$\underset{\sim}{x}^{(t+1)} \stackrel{\text{def}}{=} \underset{\sim}{x}^{(t)} + s^{(t)} \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^{(t)}).$$

Example 0.2 (Gradient ascent is gradient descent on the negative function)**.** Gradient ascent on f takes the same steps as gradient descent on $-f$, since $\frac{\partial}{\partial x}(-f) = -\frac{\partial}{\partial x} f$. In one variable, for $f(x) = -x^2$ the derivative is $f'(x) = -2x$, so gradient ascent with fixed step length s is $x^{(t+1)} = (1 - 2s)\, x^{(t)}$, the iteration of Example 0.1, which tends to the maximizer 0 when $0 < s < 1$.

Definition 0.3 (Line search)**.** In gradient descent (Definition 0.1), a **line search** [2] is a rule that chooses the step length $s^{(t)}$ by examining the values of f along the line $\left\{\underset{\sim}{x}^{(t)} - s \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^{(t)}) : s > 0\right\}$.

Definition 0.4 (Exact line search)**. Exact line search** is the line search (Definition 0.3) that chooses $s^{(t)}$ to minimize $f(\underset{\sim}{x}^{(t)} - s \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}^{(t)}))$ over $s > 0$.

Example 0.3 (Exact line search on a parabola)**.** For $f(x) = x^2$ and $x^{(t)} = 2$, the derivative is $f'(2) = 4$, so the line is $\{2 - 4s : s > 0\}$ and the values along it are

$$\begin{aligned} \phi(s) &= f(2 - 4s) \quad \text{(the value at step length } s\text{)} \\ &= (2 - 4s)^2 \quad (f(x) = x^2) \end{aligned}$$

$\phi(s) \geq 0$ for every s, and

$$\begin{aligned} \phi(0.5) &= (2 - 2)^2 \\ &= 0, \end{aligned}$$

so exact line search chooses $s^{(t)} = 0.5$, which lands on the minimizer $x^{(t+1)} = 0$ in one step.

Definition 0.5 (Descent direction)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ have a gradient $\underset{\sim}{g} \stackrel{\text{def}}{=} \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x})$ at $\underset{\sim}{x}$. A direction $\underset{\sim}{d}$ (direction) is a **descent direction** for f at $\underset{\sim}{x}$ if $\underset{\sim}{g} \cdot \underset{\sim}{d} < 0$. When $\underset{\sim}{g} \neq \underset{\sim}{0}_{p \times 1}$, $-\underset{\sim}{g}$ is a descent direction, since $\underset{\sim}{g} \cdot (-\underset{\sim}{g}) = -\|\underset{\sim}{g}\|^2 < 0$.

Example 0.4 (Descent directions for $x_1^2 + x_2^2$ at $(1,0)^\top$)**.** Let $f(\underset{\sim}{x}) = x_1^2 + x_2^2$ and $\underset{\sim}{x} = (1,0)^\top$, so

$$\begin{aligned}\underset{\sim}{g} &= (2x_1,\ 2x_2)^\top \\ &= (2,0)^\top.\end{aligned}$$

- $\underset{\sim}{d} = (-1,1)^\top$ is a descent direction:

$$\begin{aligned}\underset{\sim}{g} \cdot \underset{\sim}{d} &= -2 + 0 \\ &= -2 \\ &< 0.\end{aligned}$$

- $\underset{\sim}{d} = (0,1)^\top$ is not: $\underset{\sim}{g} \cdot \underset{\sim}{d} = 0$.
- $-\underset{\sim}{g} = (-2,0)^\top$ is: $\underset{\sim}{g} \cdot (-\underset{\sim}{g}) = -4 < 0$.

Theorem 0.1 (A short enough step downhill decreases the function)**.** *Let $f : \mathbb{R}^p \to \mathbb{R}$ have first partial derivatives that are continuous on $\mathbb{R}^p$ (continuity), and let $\underset{\sim}{x} \in \mathbb{R}^p$ be a point with $\underset{\sim}{g} \overset{def}{=} \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}) \neq \underset{\sim}{0}_{p\times 1}$. Then there is an $\bar{s} > 0$ with $f(\underset{\sim}{x} - s\underset{\sim}{g}) < f(\underset{\sim}{x})$ for every $s \in (0, \bar{s})$: a short enough step in the descent direction $-\underset{\sim}{g}$ (Definition 0.5) decreases f.*

Proof

Proof. Let $h(s) \overset{\text{def}}{=} f(\underset{\sim}{x} - s\underset{\sim}{g})$. The inner function $s \mapsto \underset{\sim}{x} - s\underset{\sim}{g}$ has derivative $-\underset{\sim}{g}^\top$ (vector-valued derivative), so by the vector chain rule (vector chain rule)

$$\begin{aligned}
h'(s) &= -\underset{\sim}{g}^\top \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x} - s\underset{\sim}{g}), && \text{(chain rule)} \\
h'(0) &= -\underset{\sim}{g}^\top \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}) && \text{(set } s = 0\text{)} \\
&= -\underset{\sim}{g} \cdot \underset{\sim}{g} && \text{(definition of } \underset{\sim}{g}\text{)} \\
&= -\|\underset{\sim}{g}\|^2 && \text{(squared length)} \\
&< 0. && (\underset{\sim}{g} \neq \underset{\sim}{0}_{p\times 1})
\end{aligned}$$

Since $h'(0) = \lim_{s\to 0}(h(s) - h(0))/s$ is negative, taking $\varepsilon = |h'(0)|$ in the definition of the limit gives an $\bar{s} > 0$ such that $(h(s) - h(0))/s < h'(0) + |h'(0)| = 0$ for $0 < s < \bar{s}$. Multiplying by $s > 0$, $h(s) < h(0)$, that is, $f(\underset{\sim}{x} - s\underset{\sim}{g}) < f(\underset{\sim}{x})$. □

Example 0.5 (How short is short enough)**.** In Example 0.1, $f(x) = x^2$ at $x \neq 0$ has $g = 2x$, and $f(x - 2sx) = (1-2s)^2x^2 < x^2$ exactly when $|1-2s| < 1$, that is, when $0 < s < 1$. So Theorem 0.1 holds with $\bar{s} = 1$, and the step $s = 1.5$ of that example was too long.

Example 0.6 (Gradient descent zigzags)**.** Let $f(\underset{\sim}{x}) = \frac{1}{2}(x_1^2 + b\,x_2^2)$ with $b > 0$, so $\frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}) = (x_1,\ bx_2)^\top$ and one gradient step of length s from $\underset{\sim}{x}$ gives $((1-s)x_1,\ (1-bs)x_2)^\top$. For exact line search (Definition 0.4), let $\phi(s) \overset{\text{def}}{=} f$ at that point:

$$\begin{aligned}
\phi(s) &= \tfrac{1}{2}((1-s)^2x_1^2 + b(1-bs)^2x_2^2), && \text{(definition of } f\text{)}\\
\phi'(s) &= -(1-s)x_1^2 - b^2(1-bs)x_2^2 && \text{(chain rule, term by term)}\\
&= -x_1^2 + s\,x_1^2 - b^2x_2^2 + b^3s\,x_2^2 && \text{(expand the products)}\\
&= s\,(x_1^2 + b^3x_2^2) - (x_1^2 + b^2x_2^2). && \text{(collect the terms in } s\text{)}
\end{aligned}$$

For $\underset{\sim}{x} \neq \underset{\sim}{0}_{2\times 1}$, $x_1^2 + b^3x_2^2 > 0$, so ϕ' is increasing in s with a single zero, and ϕ is minimized at $s^* = \dfrac{x_1^2 + b^2x_2^2}{x_1^2 + b^3x_2^2}$. From $\underset{\sim}{x}^{(0)} = (b, 1)^\top$:

$$\begin{aligned}
s^{(0)} &= \frac{b^2 + b^2}{b^2 + b^3} && \text{(substitute } x_1 = b, x_2 = 1\text{)}\\
&= \frac{2}{1+b}, && \text{(cancel } b^2\text{)}
\end{aligned}$$

and, with $r \overset{\text{def}}{=} \frac{b-1}{b+1}$,

$$\begin{aligned}
1 - s^{(0)} &= \frac{1+b-2}{1+b}\\
&= r, && \text{(common denominator)}\\
1 - b\,s^{(0)} &= \frac{1+b-2b}{1+b}\\
&= -r, && \text{(common denominator)}\\
\underset{\sim}{x}^{(1)} &= (r\,b,\ -r\cdot 1)^\top\\
&= r\,(b, -1)^\top. && \text{(the gradient step from } (b,1)^\top\text{)}
\end{aligned}$$

For $b \neq 1$ the same holds at every step, by induction (when $b = 1$, $r = 0$ and $\underset{\sim}{x}^{(1)}$ is already the minimizer): if $\underset{\sim}{x}^{(t)} = (c\,b,\ \pm c)^\top$ for a number $c \neq 0$, then s^*, which depends only on $x_1^2 = c^2b^2$ and $x_2^2 = c^2$, is again $\frac{2}{1+b}$ (the factor c^2 cancels), and the step multiplies the first entry by r and the second by $-r$. So

$$\begin{aligned} \underset{\sim}{x}^{(t)} &= (b\,r^t,\ (-r)^t)^\top, \\ f(\underset{\sim}{x}^{(t)}) &= \tfrac{1}{2}(b^2+b)\,r^{2t} \\ &= r^{2t} f(\underset{\sim}{x}^{(0)}). \end{aligned}$$

For $b \neq 1$, exactly one entry changes sign at every step: the second when $b > 1$ (then $r > 0$), and the first when $b < 1$ (then $r < 0$). So the iterates zigzag toward the minimizer $\underset{\sim}{0}_{2\times 1}$.

- With $b = 3$, $r = \frac{1}{2}$: $\underset{\sim}{x}^{(1)} = (1.5, -0.5)^\top$ and $\underset{\sim}{x}^{(2)} = (0.75, 0.25)^\top$, and f falls from 6 to 1.5 to 0.375, a quarter at each step.
- With $b = 1$, $r = 0$ and one step reaches the minimizer.
- With $b = 0.01$, $r^2 = (0.99/1.01)^2 \approx 0.96$, so f shrinks by only about 4% per step. Newton's method reaches the minimizer of any such f in one step (Theorem 2 in Newton's Method).

Definition 0.6 (Rosenbrock function)**.** The **Rosenbrock function** on $\mathbb{R}^2$ is

$$f(\underset{\sim}{x}) \stackrel{\text{def}}{=} 100(x_2 - x_1^2)^2 + (1 - x_1)^2, \qquad \underset{\sim}{x} = (x_1, x_2)^\top.$$

It is a standard test problem for optimizers [3].

Example 0.7 (Values of the Rosenbrock function)**.** We evaluate the Rosenbrock function at the minimizer $(1, 1)$, at the origin, and at the standard starting point $(-1.2, 1)$. Then we evaluate it along the curve $x_2 = x_1^2$, which is the floor of a narrow, curved valley.

R

```r
rosenbrock_rf <- function(x1, x2) 100 * (x2 - x1^2)^2 + (1 - x1)^2

points_rf <- list(minimizer = c(1, 1), origin = c(0, 0), start = c(-1.2, 1))
sapply(points_rf, function(p) round(rosenbrock_rf(p[1], p[2]), 3))
#> minimizer    origin     start
#>       0.0       1.0      24.2

# On the curve x2 = x1^2 the first term vanishes, leaving (1 - x1)^2.
x_curve_rf <- c(-1, 0, 0.5, 1, 2)
rbind(
  on_curve = rosenbrock_rf(x_curve_rf, x_curve_rf^2),
  formula = (1 - x_curve_rf)^2
)
#>          [,1] [,2] [,3] [,4] [,5]
#> on_curve    4    1 0.25    0    1
#> formula     4    1 0.25    0    1
```

Python

```python
import numpy as np

def rosenbrock_rf(x1, x2):
    return 100 * (x2 - x1 ** 2) ** 2 + (1 - x1) ** 2

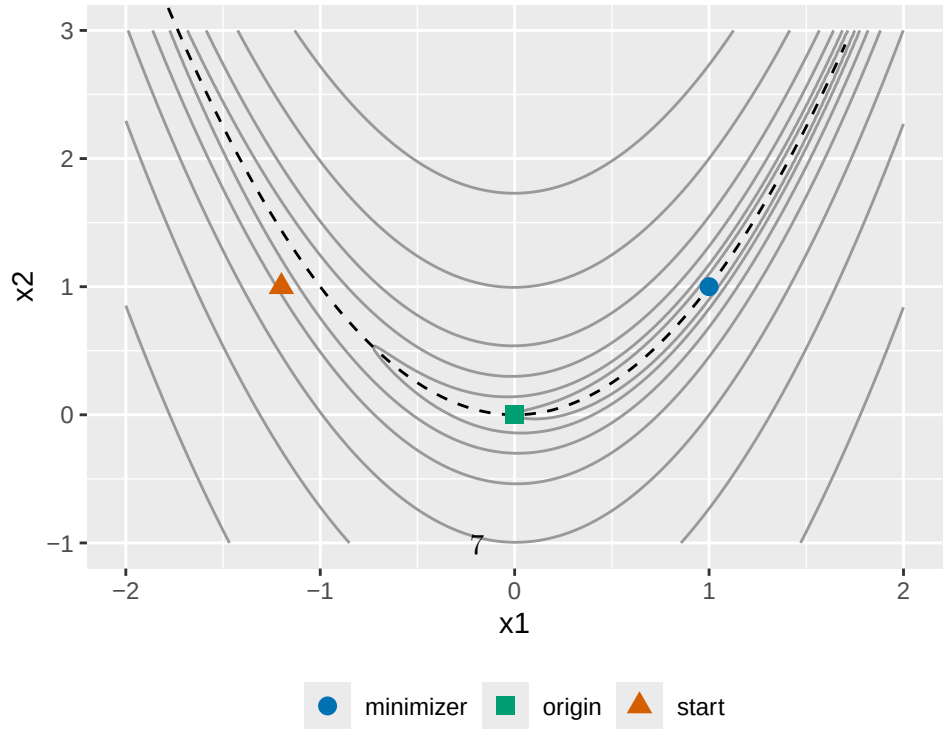
points_rf = {"minimizer": (1, 1), "origin": (0, 0), "start": (-1.2, 1)}
print({name: round(rosenbrock_rf(*p), 3) for name, p in points_rf.items()})
#> {'minimizer': 0, 'origin': 1, 'start': 24.2}

# On the curve x2 = x1^2 the first term vanishes, leaving (1 - x1)^2.
x_curve_rf = np.array([-1.0, 0.0, 0.5, 1.0, 2.0])
print("on_curve:", rosenbrock_rf(x_curve_rf, x_curve_rf ** 2))
#> on_curve: [4.   1.   0.25 0.   1.  ]
print("formula: ", (1 - x_curve_rf) ** 2)
#> formula:  [4.   1.   0.25 0.   1.  ]
```

The function is 0 at $(1, 1)$ and positive everywhere else, because both of its parts are non-negative and vanish together only when $x_1 = 1$ and $x_2 = 1$. So $(1, 1)$ is the global minimizer. At the starting point the value is 24.2, mostly from the first term, because $(-1.2, 1)$ is off the curve $x_2 = x_1^2$. On the curve the function is $(1 - x_1)^2$, which varies slowly compared with the steep walls.

R

```
library(ggplot2)
rf_grid <- expand.grid(
  x1 = seq(-2, 2, length.out = 201),
  x2 = seq(-1, 3, length.out = 201)
)
rf_grid$f <- rosenbrock_rf(rf_grid$x1, rf_grid$x2)
rf_points <- data.frame(
  name = names(points_rf),
  x1 = sapply(points_rf, `[`, 1),
  x2 = sapply(points_rf, `[`, 2)
)
rf_curve <- data.frame(x1 = seq(-2, 1.7, length.out = 200))
rf_curve$x2 <- rf_curve$x1^2
ggplot(rf_grid, aes(x1, x2)) +
  geom_contour(
    aes(z = f),
    colour = "grey60",
    breaks = c(1, 3, 10, 30, 100, 300, 1000)
  ) +
  geom_line(data = rf_curve, linetype = "dashed") +
  geom_point(aes(colour = name, shape = name), data = rf_points, size = 3) +
  scale_colour_manual(
    values = c(minimizer = "#0072B2", origin = "#009E73", start = "#D55E00")
  ) +
  scale_shape_manual(values = c(minimizer = 16, origin = 15, start = 17)) +
  coord_cartesian(xlim = c(-2, 2), ylim = c(-1, 3)) +
  labs(x = "x1", y = "x2", colour = NULL, shape = NULL) +
  theme(legend.position = "bottom")
```



Definition 0.7 (Basin of attraction)**.** For given step lengths, the **basin of attraction** of a local minimizer $\underset{\sim}{x}^*$ under gradient descent is the set of starting points $\underset{\sim}{x}^{(0)}$ from which the iterates converge to $\underset{\sim}{x}^*$ [4].

Example 0.8 (Basins of a double-well function)**.** Let $f(x) = x^4 - 2x^2$, so $\frac{\partial}{\partial x} f(x) = 4x(x-1)(x+1)$ and the stationary points are -1, 0 and 1. Both -1 and 1 are local minimizers, with $f(\pm 1) = -1$, and 0 is a local maximizer. We run gradient descent with the fixed step length $s = 0.05$ for 200 steps from the starting points -1.5, -0.5, -0.1, 0.1, 0.5, 1.5 and 3, and for three steps from $x^{(0)} = 4$.

R

```r
basin_step_bs <- 0.05
basin_starts_bs <- c(-1.5, -0.5, -0.1, 0.1, 0.5, 1.5, 3)

# 200 steps from every starting point at once.
basin_final_bs <- basin_starts_bs
for (i in 1:200) {
  basin_slope_bs <- 4 * basin_final_bs^3 - 4 * basin_final_bs
  basin_final_bs <- basin_final_bs - basin_step_bs * basin_slope_bs
}
basin_end_bs <- signif(basin_final_bs, 3)
tibble::tibble(start = basin_starts_bs, end = basin_end_bs)
#> # A tibble: 7 x 2
#>   start   end
#>   <dbl> <dbl>
#> 1  -1.5    -1
#> 2  -0.5    -1
#> 3  -0.1    -1
#> 4   0.1     1
#> 5   0.5     1
#> 6   1.5     1
#> 7   3      -1

# The first step from 3.
basin_from_three_bs <- 3 - basin_step_bs * (4 * 3^3 - 4 * 3)

# Three steps from 4: the iterates grow in size.
basin_far_bs <- numeric(4)
basin_far_bs[1] <- 4
for (i in 1:3) {
  basin_slope_bs <- 4 * basin_far_bs[i]^3 - 4 * basin_far_bs[i]
  basin_far_bs[i + 1] <- basin_far_bs[i] - basin_step_bs * basin_slope_bs
}
signif(basin_far_bs, 3)
#> [1]       4.0      -8.0      92.8 -160000.0
```

Python

```
basin_step_bs = 0.05

def basin_next_bs(x):
    return x - basin_step_bs * (4 * x ** 3 - 4 * x)

def basin_gd_bs(x0, n_steps=200):
    x = x0
    for _ in range(n_steps):
        x = basin_next_bs(x)
    return x

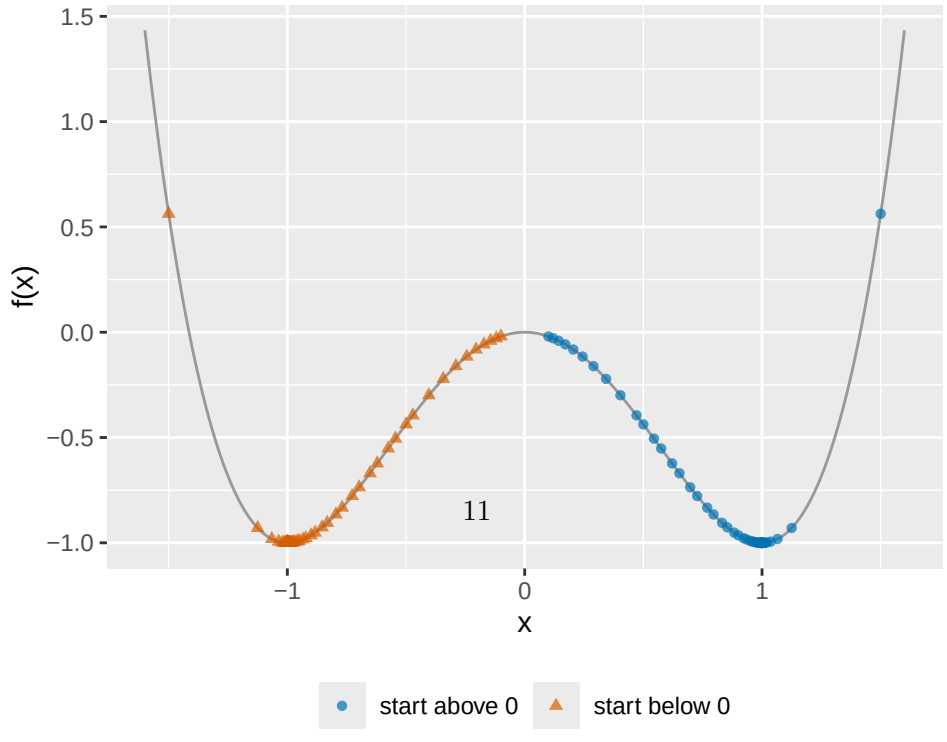
basin_starts_bs = [-1.5, -0.5, -0.1, 0.1, 0.5, 1.5, 3]
for start in basin_starts_bs:
    print(start, float(f"{basin_gd_bs(start):.3g}"))
#> -1.5 -1.0
#> -0.5 -1.0
#> -0.1 -1.0
#> 0.1 1.0
#> 0.5 1.0
#> 1.5 1.0
#> 3 -1.0

# Three steps from 4: the iterates grow in size.
basin_far_bs = [4.0]
for _ in range(3):
    basin_far_bs.append(basin_next_bs(basin_far_bs[-1]))
print([float(f"{v:.3g}") for v in basin_far_bs])
#> [4.0, -8.0, 92.8, -160000.0]
```

Each of the six starts between -1.5 and 1.5 ends at -1 or 1, whichever lies on its own side of 0, so for these starts the sign of the starting point decides which of the two the run reaches. A run from 0 itself never moves, because $\frac{\partial}{\partial x} f(0) = 0$. The basin of attraction of 1 is not all of $(0, \infty)$: the first step from $x^{(0)} = 3$ lands at -1.8, on the other side of 0, and the run ends at -1. From $x^{(0)} = 4$ the iterates, starting with $x^{(0)}$ itself, are 4, -8, 92.8 and -1.6×10^5, which grow in absolute value at every step shown. Both of these starts are far from 0, where the derivative is large and so the step $s \frac{\partial}{\partial x} f(x)$ is long. From $x^{(0)} = 3$ that step is $96\,s$ long, so the first step stays on the positive side of 0 only when s is below 0.03125. So the basin of 1 depends on s, and with $s = 0.05$ the basin is not a half-line.

R

```
# Uses basin_step_bs and basin_starts_bs from the example above.
basin_runs_bf <- matrix(0, nrow = 31, ncol = 6)
basin_runs_bf[1, ] <- basin_starts_bs[1:6]
for (i in 1:30) {
  basin_prev_bf <- basin_runs_bf[i, ]
  basin_slope_bf <- 4 * basin_prev_bf^3 - 4 * basin_prev_bf
  basin_runs_bf[i + 1, ] <- basin_prev_bf - basin_step_bs * basin_slope_bf
}
basin_side_bf <- ifelse(
  basin_starts_bs[1:6] > 0, "start above 0", "start below 0"
)
basin_x_bf <- as.vector(basin_runs_bf)
basin_side_all_bf <- rep(basin_side_bf, each = 31)
basin_paths_bf <- tibble::tibble(x = basin_x_bf, side = basin_side_all_bf)
basin_paths_bf$cost <- basin_paths_bf$x^4 - 2 * basin_paths_bf$x^2
basin_grid_bf <- seq(-1.6, 1.6, length.out = 321)
basin_curve_bf <- tibble::tibble(
  x = basin_grid_bf, cost = basin_grid_bf^4 - 2 * basin_grid_bf^2
)
basin_colours_bf <- c("start above 0" = "#0072B2", "start below 0" = "#D55E00")
ggplot2::ggplot(basin_curve_bf, ggplot2::aes(x, cost)) +
  ggplot2::geom_line(colour = "grey60") +
  ggplot2::geom_point(
    data = basin_paths_bf,
    ggplot2::aes(colour = side, shape = side), alpha = 0.7
  ) +
  ggplot2::scale_colour_manual(values = basin_colours_bf) +
  ggplot2::labs(x = "x", y = "f(x)", colour = NULL, shape = NULL) +
  ggplot2::theme(legend.position = "bottom")
```



Proposition 0.1 (Divergence of gradient descent on the double-well function)**.** *Let* $f(x) = x^4 - 2x^2$*, and run gradient descent with a fixed step length* $s > 0$*. If* $x^{(0)}$ *satisfies*

$$x^{(0)^2} > 1 + \frac{1}{2s},$$

then $\left|x^{(t)}\right| \to \infty$ *as* $t \to \infty$*.*

Proof

Proof. The derivative is $\frac{\partial}{\partial x} f(x) = 4x^3 - 4x$, so one step of gradient descent gives

$$\begin{aligned} x^{(t+1)} &= x^{(t)} - s\left(4x^{(t)^3} - 4x^{(t)}\right) && \text{(gradient descent step)} \\ &= -x^{(t)}\left(4sx^{(t)^2} - 4s - 1\right). && \text{(factor out } -x^{(t)}\text{)} \end{aligned}$$

Write $m(x) \stackrel{\text{def}}{=} 4sx^2 - 4s - 1$. Then $\left|x^{(t+1)}\right| = \left|x^{(t)}\right| m(x^{(t)})$ whenever $m(x^{(t)}) \geq 0$. Now $m(x) > 1$ exactly when $x^2 > 1 + 1/(2s)$, and $m(x)$ increases with x^2. So if $x^{(t)}$ satisfies the condition, then $\left|x^{(t+1)}\right| > \left|x^{(t)}\right|$, and $x^{(t+1)}$ satisfies it too. By induction on t, every iterate satisfies it. Since $\left|x^{(t)}\right|$ increases at every step, $x^{(t)^2} \geq x^{(0)^2}$, and so $m(x^{(t)}) \geq m(x^{(0)}) > 1$. Hence $\left|x^{(t)}\right| \geq m(x^{(0)})^t \left|x^{(0)}\right|$, which tends to ∞. □

Corollary 0.1 (Basins of the double-well function are bounded)**.** *Let* $f(x) = x^4 - 2x^2$*, and run gradient descent with a fixed step length* $s > 0$*. Every start* $x^{(0)}$ *in the basin of attraction of a local minimizer of* f *satisfies*

$$x^{(0)^2} \leq 1 + \frac{1}{2s}.$$

Proof

Proof. The iterates from a start in the basin converge to a local minimizer, so they do not diverge to ∞. By Proposition 0.1, the start therefore does not satisfy $x^{(0)^2} > 1 + 1/(2s)$. □

Example 0.9 (Divergent starts on the double-well function for three step lengths)**.** For $f(x) = x^4 - 2x^2$, Proposition 0.1 says that gradient descent diverges to ∞ from every start with $\left|x^{(0)}\right|$ above $\sqrt{1 + 1/(2s)}$. This bound is 7.14 for $s = 0.01$, 3.32 for $s = 0.05$, and 2.45 for $s = 0.1$. The derivative $4x^3 - 4x$ is unbounded, as is the second derivative $12x^2 - 4$, so the step $s\frac{\partial}{\partial x} f(x)$ grows without bound with x, and a far start overshoots. By Proposition 0.1, every fixed step length fails to converge from some starts. A larger step length lowers the bound, so more starts are guaranteed to diverge. By Corollary 0.1,

each basin of attraction lies within the same bounded interval, which narrows as s grows. In the earlier basin example, $s = 0.05$, and the start 4 lies above the bound 3.32, which is why its iterates grow.

Definition 0.8 (Plateau)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ be a function with a gradient at every point of a region, and let $\epsilon > 0$ be a tolerance. The region is a **plateau** of f if the Euclidean norm of the gradient satisfies

$$0 < \left\| \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}) \right\| < \epsilon$$

at every point $\underset{\sim}{x}$ of the region. The function is then nearly flat on the region, and the region contains no stationary point.

Example 0.10 (A cost with no stationary point)**.**
Let $f(x) = \log(1 + \mathrm{e}^{-x})$. Its derivative is $\frac{\partial}{\partial x} f(x) = -1/(1 + \mathrm{e}^{x})$, which is negative for every x, so f has no stationary point at all.
At $x = 0$ the gradient has size 0.5. At $x = 10$ it has size 4.54×10^{-5}, which is small but not zero. A step of gradient descent subtracts the gradient times the step length. The gradient is negative, so the iterates keep moving to the right, more and more slowly. With tolerance $\epsilon = 10^{-4}$, the region $x \geq 10$ is a plateau (Definition 0.8). A run that enters this plateau has not converged; the run has only slowed down.

Definition 0.9 (Armijo condition)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ have first partial derivatives, let $\underset{\sim}{x} \in \mathbb{R}^p$, and let $g \overset{\text{def}}{=} \frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x})$. A step length $s > 0$ satisfies the **Armijo condition** [5] with constant $c \in (0, 1)$ at $\underset{\sim}{x}$ if

$$f(\underset{\sim}{x} - s\underset{\sim}{g}) \leq f(\underset{\sim}{x}) - c\, s\, \|\underset{\sim}{g}\|^2.$$

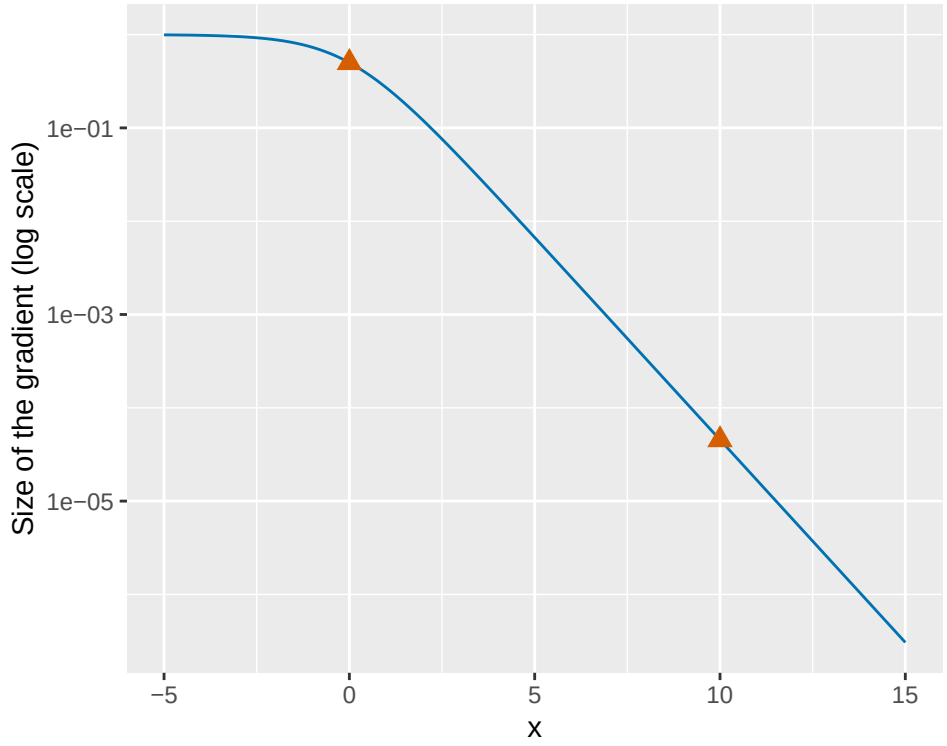
Example 0.11 (The Armijo condition on a parabola)**.** Let $f(x) = x^2$ and $x = 2$, so $g = f'(2) = 4$ and $\|g\|^2 = 16$. With $c = 1/4$, the condition on a step length $s > 0$ is

$$\begin{aligned} (2 - 4s)^2 &\leq 4 - 4s && (f(2 - 4s) \leq f(2) - c\, s\, \|g\|^2) \\ 4 - 16s + 16s^2 &\leq 4 - 4s && (\text{expand the square}) \\ 16s^2 &\leq 12s && (\text{add } 16s - 4 \text{ to both sides}) \\ s &\leq 3/4. && (\text{divide by } 16s > 0) \end{aligned}$$

So $s = 1/2$ satisfies the condition: $f(0) = 0 \leq 4 - 2 = 2$. And $s = 1$ does not: $f(-2) = 4 > 4 - 4 = 0$.

R

```
plateau_grid_pf <- seq(-5, 15, length.out = 401)
plateau_exp_pf <- exp(plateau_grid_pf)
plateau_size_pf <- 1 / (1 + plateau_exp_pf)
plateau_curve_pf <- tibble::tibble(x = plateau_grid_pf, size = plateau_size_pf)
plateau_marks_x_pf <- c(0, 10)
plateau_marks_exp_pf <- exp(plateau_marks_x_pf)
plateau_marks_size_pf <- 1 / (1 + plateau_marks_exp_pf)
plateau_marks_pf <- tibble::tibble(
  x = plateau_marks_x_pf, size = plateau_marks_size_pf
)
plateau_aes_pf <- ggplot2::aes(x, size)
ggplot2::ggplot(plateau_curve_pf, plateau_aes_pf) +
  ggplot2::geom_line(colour = "#0072B2") +
  ggplot2::geom_point(
    data = plateau_marks_pf, colour = "#D55E00", shape = 17, size = 3
  ) +
  ggplot2::scale_y_log10() +
  ggplot2::labs(x = "x", y = "Size of the gradient (log scale)")
```



Python

```
import matplotlib.pyplot as plt
import numpy as np

plateau_grid_pf = np.linspace(-5, 15, 401)
plateau_marks_pf = np.array([0, 10])
fig, ax = plt.subplots(figsize=(5, 4), layout="constrained")
ax.plot(plateau_grid_pf, 1 / (1 + np.exp(plateau_grid_pf)), color="#0072B2")
#> [<matplotlib.lines.Line2D object at 0x7fc44e08f250>]
```

Definition 0.10 (Backtracking line search)**.** Let $s_0 > 0$ be an initial step length, $\beta \in (0,1)$ a shrink factor, and $c \in (0,1)$. **Backtracking line search** [5] is the line search that chooses $s^{(t)}$ to be the largest of $s_0, \beta s_0, \beta^2 s_0, \ldots$ that satisfies the Armijo condition with constant c at $\underset{\sim}{x}^{(t)}$.

Example 0.12 (Backtracking on a parabola)**.** For $f(x) = x^2$ at $x = 2$ with $c = 1/4$, the Armijo condition holds exactly when $s \le 3/4$ (Example 0.11). With $s_0 = 1$ and $\beta = 1/2$, the trial step lengths are $1, 1/2, 1/4, \ldots$. The first, $s = 1$, exceeds $3/4$ and is rejected. The second, $s = 1/2$, is at most $3/4$ and is accepted. So backtracking chooses $s^{(0)} = 1/2$, after two evaluations of f.

Theorem 0.2 (A bound on the error of gradient descent on a quadratic)**.** *Let* $\mathbf{S}$ *be a* $p \times p$ *positive definite matrix with eigenvalues* $\lambda_1, \ldots, \lambda_p$*, let* $\underset{\sim}{c} \in \mathbb{R}^p$*, and let* $f(\underset{\sim}{x}) = \frac{1}{2}\,\underset{\sim}{x}^\top \mathbf{S}\underset{\sim}{x} - \underset{\sim}{c}\cdot\underset{\sim}{x}$*, whose unique minimizer is* $\underset{\sim}{x}^* = \mathbf{S}^{-1}\underset{\sim}{c}$ *(Theorem 2 in Newton's Method). Gradient descent with a fixed step length* $s > 0$ *from any* $\underset{\sim}{x}^{(0)}$ *has errors satisfying, for* $t = 1, 2, \ldots,$

$$\left\|\underset{\sim}{x}^{(t)} - \underset{\sim}{x}^*\right\| \le \rho(s)^t \left\|\underset{\sim}{x}^{(0)} - \underset{\sim}{x}^*\right\|, \qquad \rho(s) \overset{def}{=} \max_{1 \le i \le p} |1 - s\,\lambda_i|.$$

Proof

Proof. Let $\underset{\sim}{e}^{(t)} \overset{\text{def}}{=} \underset{\sim}{x}^{(t)} - \underset{\sim}{x}^*$ be the error at step t. The gradient of f is $\frac{\partial}{\partial \underset{\sim}{x}} f(\underset{\sim}{x}) = \mathbf{S}\underset{\sim}{x} - \underset{\sim}{c}$ (see the proof of Theorem 2 in Newton's Method), and $\underset{\sim}{c} = \mathbf{S}\underset{\sim}{x}^*$, so

$$\begin{aligned}
\underset{\sim}{e}^{(t+1)} &= \underset{\sim}{x}^{(t)} - s\,(\mathbf{S}\underset{\sim}{x}^{(t)} - \underset{\sim}{c}) - \underset{\sim}{x}^* && \text{(definition of gradient descent)} \\
&= \underset{\sim}{x}^{(t)} - s\,\mathbf{S}(\underset{\sim}{x}^{(t)} - \underset{\sim}{x}^*) - \underset{\sim}{x}^* && (\underset{\sim}{c} = \mathbf{S}\underset{\sim}{x}^*) \\
&= (\mathbf{I}_p - s\,\mathbf{S})\,\underset{\sim}{e}^{(t)}. && \text{(collect the terms in } \underset{\sim}{e}^{(t)})
\end{aligned}$$

By induction on t, $\underset{\sim}{e}^{(t)} = (\mathbf{I}_p - s\,\mathbf{S})^t\,\underset{\sim}{e}^{(0)}$. Write $\mathbf{S} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top$ (spectral theorem). Then $\mathbf{I}_p - s\,\mathbf{S} = \mathbf{Q}(\mathbf{I}_p - s\,\mathbf{\Lambda})\mathbf{Q}^\top$ because $\mathbf{Q}\mathbf{Q}^\top = \mathbf{I}_p$, and the tth power cancels each inner $\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}_p$, so

$$\begin{aligned}
\left\|\underset{\sim}{e}^{(t)}\right\| &= \left\|\mathbf{Q}(\mathbf{I}_p - s\,\mathbf{\Lambda})^t\mathbf{Q}^\top\underset{\sim}{e}^{(0)}\right\| && \text{(the factorization above)} \\
&= \left\|(\mathbf{I}_p - s\,\mathbf{\Lambda})^t\mathbf{Q}^\top\underset{\sim}{e}^{(0)}\right\| && \text{(orthogonal matrices preserve length)} \\
&\le \rho(s)^t \left\|\mathbf{Q}^\top\underset{\sim}{e}^{(0)}\right\| && \text{(each coordinate is multiplied by } (1 - s\lambda_i)^t) \\
&= \rho(s)^t \left\|\underset{\sim}{e}^{(0)}\right\|. && \text{(orthogonal matrices preserve length)}
\end{aligned}$$

The length-preserving steps apply orthogonal matrices preserve length. The inequality holds because $\mathbf{I}_p - s\,\mathbf{\Lambda}$ is diagonal, so its tth power multiplies coordinate i by $(1 - s\,\lambda_i)^t$, whose absolute value is at most $\rho(s)^t$. □

Example 0.13 (The error multiplier for a step length of one quarter)**.** Let $\mathbf{S} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ with eigenvalues 3 and 1 (eigenvalue example), and $\underset{\sim}{c} = \underset{\sim}{0}_{2\times 1}$, so $\underset{\sim}{x}^* = \underset{\sim}{0}_{2\times 1}$. With $s = 0.25$, the factors $1 - s\,\lambda_i$ are 0.25 and 0.75, so $\rho(s) = 0.75$ and

$$\left\|\underset{\sim}{x}^{(t)}\right\| \le 0.75^t \left\|\underset{\sim}{x}^{(0)}\right\|.$$

Along the eigenvector for $\lambda = 3$ the error shrinks fast, and along the eigenvector for $\lambda = 1$ it shrinks slowly; the bound follows the slower one.

Theorem 0.3 (The best constant step length)**.** *In the setting of Theorem 0.2, let* $\lambda_{\min}$ *and* $\lambda_{\max}$ *be the smallest and largest eigenvalues of* $\mathbf{S}$*. The factor* $\rho(s) = \max_i |1 - s\,\lambda_i|$ *of Theorem 0.2 is smallest over* $s > 0$ *at*

$$s^* = \frac{2}{\lambda_{\min} + \lambda_{\max}},$$

where

$$\rho(s^*) = \frac{\kappa - 1}{\kappa + 1}, \qquad \kappa = \kappa(\mathbf{S}),$$

the condition number of $\mathbf{S}$*.*

Proof

Proof. For $s > 0$, each $1 - s\,\lambda_i$ lies between $u_{\min} \overset{\text{def}}{=} 1 - s\,\lambda_{\max}$ and $u_{\max} \overset{\text{def}}{=} 1 - s\,\lambda_{\min}$, so $|1 - s\,\lambda_i| \le \max(u_{\max}, -u_{\min})$. Both $\lambda_{\min}$ and $\lambda_{\max}$ occur among the λ_i, and $|u_{\max}| \ge u_{\max}$ and $|u_{\min}| \ge -u_{\min}$, so the maximum is attained:

$$\rho(s) = \max(1 - s\,\lambda_{\min},\ s\,\lambda_{\max} - 1).$$

The first expression is strictly decreasing in s, and the second is strictly increasing, and they coincide when

$$\begin{aligned} 1 - s\,\lambda_{\min} &= s\,\lambda_{\max} - 1 && \text{(set the two expressions equal)} \\ 2 &= s\,(\lambda_{\min} + \lambda_{\max}) && \text{(collect the terms in } s\text{)} \\ s &= \frac{2}{\lambda_{\min} + \lambda_{\max}} = s^*. && \text{(divide)} \end{aligned}$$

For $s < s^*$, $\rho(s)$ is at least the decreasing expression, which exceeds its value at s^*. For $s > s^*$, $\rho(s)$ is at least the increasing expression, which exceeds its value at s^*. So s^* is the minimizer, and

$$\begin{aligned}
\rho(s^*) &= 1 - \frac{2\,\lambda_{\min}}{\lambda_{\min} + \lambda_{\max}} && \text{(first expression at } s^*\text{)} \\
&= \frac{\lambda_{\max} - \lambda_{\min}}{\lambda_{\max} + \lambda_{\min}} && \text{(common denominator)} \\
&= \frac{\kappa - 1}{\kappa + 1}. && \text{(divide the numerator and denominator by } \lambda_{\min}\text{)}
\end{aligned}$$

□

Example 0.14 (The best step length for eigenvalues 3 and 1)**.** For $\mathbf{S} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ with eigenvalues 3 and 1, Theorem 0.3 gives $s^* = 2/(3+1) = 0.5$, $\kappa = 3$ (condition number example), and $\rho(s^*) = (3-1)/(3+1) = 0.5$. A grid search over $s = 0.01, 0.02, \ldots, 1$ finds the smallest $\rho(s)$ at $s = 0.5$, the same step length. Each step then multiplies the error by at most 0.5.

Exercise 0.1 (Which step lengths shrink the error bound)**.** In the setting of Theorem 0.2, find all $s > 0$ with $\rho(s) < 1$, and write the answer as a condition on s that involves $\lambda_{\max}$.

Solution

Solution 0.1. $\rho(s) < 1$ says that $|1 - s\,\lambda_i| < 1$ for every i, which is $-1 < 1 - s\,\lambda_i < 1$. The right inequality holds for every $s > 0$ because $\lambda_i > 0$. The left inequality is $s\,\lambda_i < 2$, that is, $s < 2/\lambda_i$. These p inequalities, one for each eigenvalue, all hold exactly when the inequality for the largest eigenvalue holds, since it is the most restrictive:

$$0 < s < \frac{2}{\lambda_{\max}}.$$

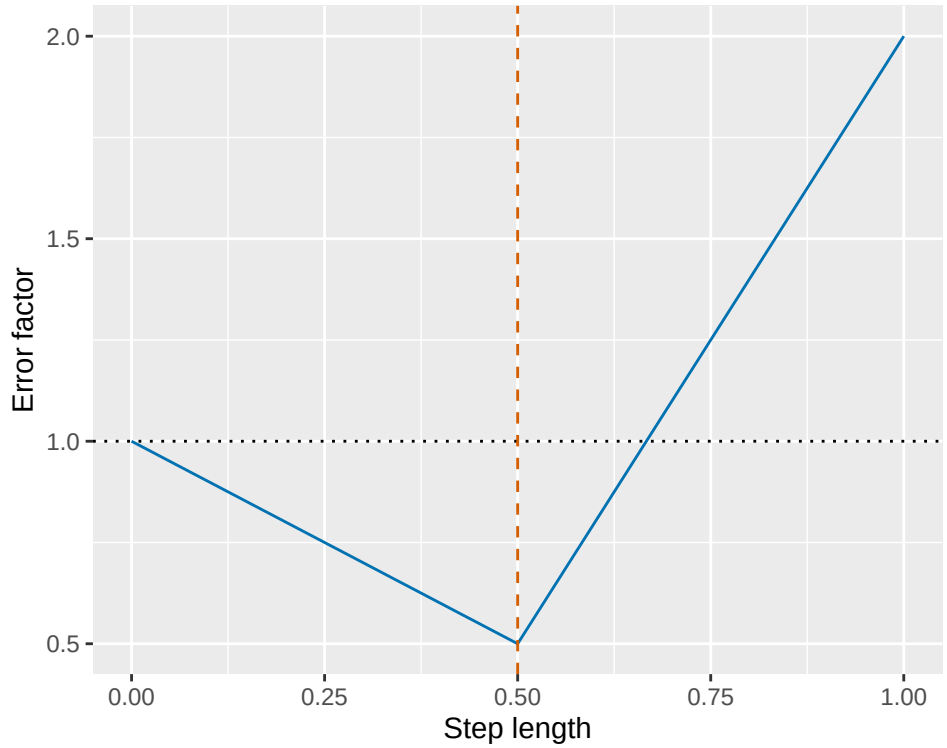
Exercise 0.2 (The best step length for eigenvalues 1 and 9)**.** In gradient descent, a fixed step length that is too short converges slowly, and one that is too long diverges. Theorem 0.3 gives the best one from the eigenvalues of $\mathbf{S}$.
Let $\mathbf{S} = \begin{bmatrix} 1 & 0 \\ 0 & 9 \end{bmatrix}$. Using Theorem 0.3, find s^*.

Solution

Solution 0.2. $\mathbf{S}$ is diagonal, so its eigenvalues are its diagonal entries 1 and 9. Then

R

```r
bcs_grid_pf <- seq(0, 1, length.out = 401)
bcs_first_pf <- abs(1 - 3 * bcs_grid_pf)
bcs_second_pf <- abs(1 - bcs_grid_pf)
bcs_rho_pf <- pmax(bcs_first_pf, bcs_second_pf)
bcs_curve_pf <- tibble::tibble(s = bcs_grid_pf, rho = bcs_rho_pf)
bcs_aes_pf <- ggplot2::aes(s, rho)
ggplot2::ggplot(bcs_curve_pf, bcs_aes_pf) +
  ggplot2::geom_line(colour = "#0072B2") +
  ggplot2::geom_hline(yintercept = 1, linetype = "dotted") +
  ggplot2::geom_vline(
    xintercept = 0.5, colour = "#D55E00", linetype = "dashed"
  ) +
  ggplot2::labs(x = "Step length", y = "Error factor")
```



Python

```python
import matplotlib.pyplot as plt
import numpy as np

bcs_grid_pf = np.linspace(0, 1, 401)
bcs_rho_pf = np.maximum(np.abs(1 - 3 * bcs_grid_pf), np.abs(1 - bcs_grid_pf))
fig, ax = plt.subplots(figsize=(5, 4), layout="constrained")
ax.plot(bcs_grid_pf, bcs_rho_pf, color="#0072B2")
#> [<matplotlib.lines.Line2D object at 0x7fc44be916d0>]
ax.axhline(1, color="black", linestyle=":")
#> <matplotlib.lines.Line2D object at 0x7fc44be91810>
ax.axvline(0.5, color="#D55E00", linestyle="--")
#> <matplotlib.lines.Line2D object at 0x7fc44be91950>
ax.set_xlabel("Step length")
```

$$s^* = \frac{2}{1+9} = 0.2.$$

Exercise 0.3 (The best error factor for eigenvalues 1 and 9)**.** In the setting of Exercise 0.2, find $\rho(s^*)$.

Solution

Solution 0.3. The condition number is $\kappa(\mathbf{S}) = 9/1 = 9$, so by Theorem 0.3

$$\rho(s^*) = \frac{9-1}{9+1} = 0.8.$$

Exercise 0.4 (Steps to cut the error bound by a factor of ten)**.** In the setting of Exercise 0.2 and Exercise 0.3, find the smallest $t = 1, 2, \ldots$ with $\rho(s^*)^t \leq 0.1$.

Solution

Solution 0.4. From Solution 0.3, $\rho(s^*) = 0.8$. Taking logarithms, $\rho(s^*)^t \leq 0.1$ becomes $t \ln \rho(s^*) \leq \ln 0.1$. Here $\ln \rho(s^*) < 0$, so dividing by it flips the inequality: $t \geq \ln 0.1 / \ln \rho(s^*) \approx 10.32$. The smallest such t is 11.

[1] Wikipedia contributors, "Gradient descent — Wikipedia, the free encyclopedia." 2026. Available: https://en.wikipedia.org/wiki/Gradient_descent

[2] Wikipedia contributors, "Line search — Wikipedia, the free encyclopedia." 2026. Available: https://en.wikipedia.org/wiki/Line_search

[3] Wikipedia contributors, "Rosenbrock function — Wikipedia, the free encyclopedia." 2026. Available: https://en.wikipedia.org/wiki/Rosenbrock_function

[4] Wikipedia contributors, "Attractor — Wikipedia, the free encyclopedia." 2026. Available: https://en.wikipedia.org/wiki/Attractor

[5] Wikipedia contributors, "Backtracking line search — Wikipedia, the free encyclopedia." 2026. Available: https://en.wikipedia.org/wiki/Backtracking_line_search